PL SHOP
INITIAL COMMENTS

- Baked Enamel: Clean, degrease zinc-coated metal surface; one coat zinc oxide primer sprayed and baked; two coats semi-gloss enamel sprayed and baked.

14 28 16 - Elevator Controls
Standard: Open protocol, non-proprietary, specified with trouble shooting and operational software, and serviceable by all licensed elevator contractors.

Manufacturer: Motion Control Engineering. http://www.mceinc.com/.

Elevator Keys: Standard: Provide 3 keys, installing 1 in elevator pit.

Closeout Submittal: Keys: Provide manufacturer name and model numbers for keys and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

REVIEW THESE COMMENTS TO VERIFY THEY ALL WERE PICKED UP

Division 21 - Fire Suppression

21 00 00 - Fire Suppression
Note: This document is a listing of University requirements and is not a complete specification. Consultant shall incorporate these requirements into a full technical specification.

General
Generally, the University requires that the consultant prepare a specification for a Contractor to design and construct (design-build) the fire protection systems to meet the specifications contained herein, including the various design and performance criteria delineated and to be responsible for the actual performance of the system according to these criteria. Verify with Project Manager that project will be design-build.

Quality Assurance
- Contractor to have C-16 license.
- Work shall be done in accordance with the N.F.P.A., the California Administrative Code, the Campus Standards including those of the Deputy State Fire Marshal, the California Building and Fire codes, and the requirements of the California State Fire Marshal. Plans shall be stamped by a California State licensed Fire Protection Engineer or mechanical engineer.

Submittals
- Provide standpipe and wet pipe fire sprinkler systems installation shop drawings, component submittal sheets and hydraulic calculations for approval of University representative and State Fire Marshal. University's Representative shall submit to Designated Campus Fire Marshal who is State Fire Marshal Representative. Do not begin installation until approval has been received.
- Shop drawings shall indicate standpipe and sprinkler assembly, wiring diagrams, including zone control and detection, flow, tamper switches and supervisory devices, inspector's test valve, fire department connection, post indicating valve, etc.
- Shop drawings shall indicate fire protection system connection to site water system including check assemblies.
- Contractor shall notify fire alarm subcontractor if waterflow or tamper switch or P.I.V. devices are relocated from locations shown on bid documents.

Design Criteria

- Contractor may with permission of the Designated State Fire Marshal, in lieu of the hydraulic calculations specified, use the prescriptive sizing method outlined in NFPA No. 13.
- The sprinkler system shall provide complete automatic wet-pipe sprinkler protection throughout the building including the attic space.
- The sprinkler system shall be zoned. Generally a zone should cover no more than a single floor.
- The University's Representative shall work with the Deputy State Fire Marshal to determine the Project hydraulic requirements (including occupancy classification), allowing for future expansion of system, system pressure variations, and the desired margin of safety.
- The following minimum pressure values should be used for sprinkler system design regardless of available site pressure values.
  - Static - 50 psi
  - Residual - 40 psi
  - Flow pressure - 35 psi at 1000 gpm
- Sprinklers shall be provided in combustible construction both below and above ceilings. Quick response heads shall be provided where required by the State Fire Marshal. On-off heads shall be provided in extra hazard areas.
- Backflow prevention - Conform to AWWA recommendations for protection of potable water system. Currently this consists of a single check valve assembly.
- Gongs/Bells - Use fire alarm system horns for annunciation of flow switches and monitoring of tamper switches. Do not install water gongs or bells.
- Fire protection water connections to a building shall be made with a Post Indicating Valve, check valve and fire department connection that are unique to that building. For very small clusters of buildings, an exception to this policy may be made by the Fire Marshal after receiving a written request by the Project Manager and after evaluation of the site plan.
- Access to main shutoff valves shall be through no more than one door which is located on the exterior of the building.

Commented [MDS1]: Albert needs to check these pressures

21 05 19 - Meters and Gages of Fire-Suppression Systems
Refer to CSU Building Metering Guide:
http://www.calstate.edu/cpdc/ae/gsf/documents/CSU_Metering_Guide.pdf

21 09 00 - Instrumentation and Control for Fire Suppression
Standards: Conduit: ¾-inch, minimum

21 12 00 - Fire-Suppression Standpipes
Buried piping and fittings: Cement-mortar lined, ductile iron pipe conforming to AWWA C-151, Class 200, 21/45 with Tyton or as approved, mechanical joints and fittings conforming to NFPA-24. Install pipe joints anchored with tie rods and restraints. Provide concrete thrust blocks at elbows and offsets conforming to NFPA-13 and NFPA-24. Other types of UL/FM reviewed piping including PVC C900 Class 200 may be used as approved by University's Representative and the State Fire Marshal. Thrust block installation shall be inspected by State Fire Marshal.

Commented [MDS2]: Albert to check if these work

Above ground piping and fittings: Standard weight, Schedule 40, welded and seamless black steel, ASTM A-135 or A-53 pipe conforming to NFPA-13. Fittings for pipe sizes 2-1/2 inch and smaller shall be malleable iron, black, 150 lb. standard or cast iron, 175 lb. WP, screwed ends. Fittings for pipe sizes 3 inch and larger may be as above or weld fittings, ANSI 16.9, Schedule 40, flanged or welded fittings, cast iron, 175 lb. WP. Clamp type UL/FM approved fittings may be used on concealed piping where listed by the State Fire Marshal. Piping installed outdoors shall be galvanized.

Alternate piping system to be Victaulic, Grinnel, or equal grooved piping systems.
CPVC piping may be used in locations approved by NFPA 13 and the State Fire Marshal.

Commented [MD53]: Albert to check

21 13 00 - Fire Suppression Sprinkler Systems
Preference: Dry-Pipe Sprinkler Systems:
Use where water could damage equipment in room.

Standards: Alarm Devices:
Water-Motor Operated Alarm, Electrically Operated Alarm, Water-Flow Indicator, Valve Supervisory Switch, and Indicator-Post Supervisory Switch.

Sprinkler and Valves
- Flanged 175 psi minimum water pressure rating.
- Gate (2-1/2 inch and larger): Iron body, bronze fitted 175 lb. WWP, BB, OS & Y, solid wedge flanged. Standpipe valves required shall be size 2-1/2 inch and shall have hose ends with caps and chains as required by the Deputy State Fire Marshal, bronze finish outdoors.
- Check (2-1/2 inch and larger): Iron body, bronze fitted 175 lb. WWP, bronze disc, horizontal swing, BB, flanged.
- Gate (2 inch and smaller): Bronze 175 lb. WWP, block pattem, OS & Y, solid wedge, screwed ends.
- Sprinkler Heads - Heads subject to vandalism shall be protected with wire guards. Heads installed in hazardous storage areas shall be on-off type and shall be submitted to Deputy State Fire Marshall for approval.
- Escutcheons: Install for exposed to view pipe penetrations through finished partitions, walls and floors.
- Pressure Gauges: Dial type with 0-300 lb. range, with brass case, glass face and attachment.

Backflow Prevention Assembly (where required):
BEECO-Hersey, Watts, Febco or equal, reduced pressure type assembly, UL/FM approved and City of San Luis Obispo Public Works Engineering Standards approved and listed with OS & Y gate vales and tamper switches.
Execution
- Piping runs shall be checked beforehand and with other trades to insure clearance. Provide maximum head room and run piping to maintain proper clearance for maintenance and to clear openings in exposed areas. Piping shall be run in strict coordination with mechanical ducts and equipment, structural, and architectural conditions.
- Cutting structural members for passing sprinkler pipe or supports will not be permitted except with approval from the University.
- Piping fittings shall not be located over electrical machinery or equipment, unless adequate insulating protection (16-gauge galvanized sheet metal pans with hemmed edges) is provided against drip caused by leaks.
- In general, shut-off valves shall be oriented so that the valve stem is horizontal. If valves controlling sprinkler systems are inaccessible or over an elevation of 7'-0"or greater above the finished floor or grade, install permanent ladders, clamped threads on risers, chain operators or similar devices to provide access to authorized personnel.

No butterfly valves
gate valves only

- Butterfly valves shall be installed with the position indicator located so it can easily be read, and the crank is within reach and easily operable.
- Supports - Provide seismic restraints for piping and equipment as required by NFPA and C.C.R. Title 24, whichever is stricter.
- Inspector's Test Connection - Provide test connections approximately 6 feet above the floor for the sprinkler system or portion of each sprinkler system equipped with an alarm device. Locate at the hydraulic most remote part of each system. Provide test connection piping to a location where the discharge will be readily visible and where water may be discharged without property damage. Provide discharge orifice of same size as smallest sprinkler orifice.
- Main Drain - Provide separate drain piping to discharge at set point outside building or to sight cones attached to drain of adequate size to readily receive the full flow from drain under maximum pressure. Install main drains to the outside of the building in such a manner that full flow from the drain will not damage landscaping or surroundings.
- Fire Department Connections - Provide connections approximately 3 feet above finish grade, of the University approved two-way type, with National Standard female 4" hose threads with plug, chain, and identifying fire department connection escutcheon plate. Connection shall be free standing (or wall mounted - coordinate with Fire Marshal).
- Flushing and disinfection - Disinfect potable portions of the Fire Protection System as required by the State Plumbing and Health Codes and the Deputy State Fire Marshal. Newly installed piping is to be kept isolated from the system until bacteriologically acceptable. If isolation is provided by a closed gate valve, pressure testing for leakage in the new facilities shall be conducted before bacteriological acceptance. The University shall designate method and sequence of connecting mains to minimize contamination danger.
- Tests - Tests shall be conducted at such times as directed by and in the presence of the University's Representative and the State Fire Marshal and shall be provided as required by NFPA-13, 24, and the State Fire Marshal. Piping shall be hydrostatically tested as required by NFPA-13 and NFPA-24but not less than 150 psi for two (2) hours.
- Operating parts, including valves, water flow detectors, and valve supervisory switches shall be tested for proper operation.
- Contractor to submit system certification forms per NFPA-13

Division 22 - Plumbing

22 05 00 - Common Work Results for Plumbing

Following is a brief description of the features which should be incorporated in the specifications for various plumbing materials and equipment. Where a specific product is stated with model number, that product should be used as the basis of design and listed as the first named in the specifications. Manufacturers listed without model number are intended to give the designer input on manufacturers whose products have been found to be acceptable.

General Quality Level:

Generally, plumbing material and equipment selected should be institutional grade. Long life and simple low cost maintenance are critical attributes for nearly all CAL POLY projects. Plumbing fixtures in public areas should also be selected to endure rough use and daily janitorial cleaning.

General-Duty Valves Requirements:

- Isolation Valves: Provide sufficient number of valves for ease of service, and to reduce inconvenience to users due to outages and draining of systems for minor repairs.
- Relief valves: Daylight to a conspicuous location. Plumb to sewer.

Ball valve with Stainless steel ball and stem from1/2"-1.5" and from 2" and up either bronze gate valve or epoxy coated resilient wedge iron body gate valve

Identification for Plumbing Piping and Equipment Requirements:

- Identification Charts: Location should be reviewed and approved by Project Manager in consultation with Facility Services.
- Piping Service: Identification Tags: Provide with abbreviated legend on 1st line and pipe size on 2nd line. Locate to be visible from exposed points of observation. Where 2 or more pipes run parallel, place printed legend and other markers in same relative location.
- Valves Service Identification Tags: Provide with abbreviated legend on 1st line and valve service chart number on 2nd line. Identification Charts: Provide two (2) satin finished extruded aluminum frames with rigid clear plastic glazing; 8-1/2 x 11 inches, minimum for each chart. In addition, provide electronic copy of each chart.

Keys for Cabinets and Padlocks:

- Cabinets and Equipment: Provide 2 keys per panel. Coordinate with Section 08 06 05 – Key Schedule.
- Padlocks: Coordinate with Section 08 06 05 – Key Schedule.
- Closeout Submittal: Provide panel keys separated and labeled. Provide location, room number, quantity, manufacturer name and model numbers of keys, and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

Testing

- All new piping shall be tested prior to tie in to existing systems.
- Do not test against existing valves when connecting into an existing system. Provide a slip blind at the valve flange or other suitable isolation.
- Test gauges shall have 3" minimum dial, with oil fill and gauge cock. Gauge calibration shall be verified to the satisfaction of the University's Inspector prior to commencing testing.
- Domestic and Industrial Hot & Cold Water Piping
- 150 PSIG test pressure for a period of 4 hours using a test medium of water.
- 200 PSIG test pressure for a period of 4 hours using a test medium of water for portions of the system which serve both the fire suppression system and the domestic water .
- Drain, Waste & Vent Piping Including Lab Waste: 10 feet of head for a period of 1 hour using a test medium of Water.
- Natural Gas Piping: 50 PSIG test pressure for a period of 4 hours using a test medium of air.
- Compressed Air Piping: 150 PSIG test pressure for a period of 4 hours using a test medium of air.
- Pure Water Piping Systems: 100 PSIG test pressure for a period of 4 hours using a test medium of purified water.22 05 19 Meters and Gages for Plumbing Piping
  - Water meters for domestic water should be nutating disk type sufficiently accurate to record the lowest expected flow (typically the flush of a single low flush toilet). Provide compound metering if necessary to accommodate a large range of flows.
  - Provide a permanently piped full size meter bypass on all domestic water services 3" and larger. Valving should be arranged so that the meter can be removed without a disruption of water service.
  - All water meters should record in cubic feet.
  - Meters for irrigation may be either nutating disk or turbine type depending on expected flow.

22 06 00 - Schedule for Plumbing

Boiler / Chiller Room Plumbing

Provide floor drains / sinks for relief valve drains and to allow for drain down of closed loop systems. Coordinate locations with equipment layout so that piped equipment drains will not be a trip hazard in service aisles. Select floor drains/sinks to contain spill over and splashing due to equipment drain discharge.

Provide make up water through a RP backflow preventer for each closed loop system.

Plaster / Sand Use.

Rooms where plaster or other sediment material is used shall be equipped with a dedicated above grade sand trap on the drain outlet of each sink. Typical applications include art studios using plaster, anthropology labs using plaster for bone castings, etc.

Dark Rooms

Provide RP back flow preventers for hot and cold water to dark room sinks.

Fume Hoods

For each plumbing fume hood plumbing service, provide a manual isolation valve and union at the piping drop above each fume hood so that the hood can be easily removed with minimal disruption of building piping.

Commercial Kitchens

Per City of San Luis Obispo Waste Water Discharge Ordinance, a large outdoor grease trap is typically required. Position grease trap to allow for easy access for pumping and minimal nuisance odors at occupied locations while pumping.

22 06 10 - Schedule for Plumbing Piping and Pumps

Piping: Standard: U.S.-manufactured.

22 10 00 - Plumbing Piping

All underground plumbing utility piping shall have a minimum of 12" sand encasement.
All underground plumbing utility piping shall have a minimum depth of 24" unless otherwise approved by the administrative authority after all existing utilities have been potholed by the contractor.

Design Documentation

Programming; For wet laboratories, studios, dark rooms, commercial kitchens, laundries, and other special use rooms requiring plumbing utilities to accommodate special functions, the Design Professional shall gather and document all information required to confirm Plumbing requirements as part of room programming. This information shall be included with the Design Development Submittal. Obtain and document the following information for each building room where special use plumbing services will be required.

- Room name
- Person requesting the plumbing services, Interviewer, Date
- Room use
- A list of required plumbing services. Include qualitative and quantitative data if available.

- Functions / processes which the required plumbing services are intended to accommodate. Provide sufficient information to ascertain the need for special plumbing provisions such as back flow prevention, pressure regulation, drain sediment traps, etc.
- A list of all equipment / apparatus which will require plumbing services. Include equipment manufacturer's data (connection size, use rate, etc.) if available.
- The above programming information may be omitted for rest rooms and residential kitchens where all required design data is covered by applicable codes.

Calculations:
The Design Professional shall include final plumbing calculations with the 50% Working Drawing submittal. These calculations shall include the following:

- The room programming documents covered above.
- All assumptions clearly stated. In particular, document assumptions that had to be made in order to proceed with the design in the case of insufficient data being provided by the users. The intent here is to flag assumptions which the University should verify as correct.
- Applicable code requirements
- Tabulated design criteria for each plumbing service by system including:
  - System
  - Location of Service (Room number)
  - Required flow quantity for each location including: water use fixture units, drainage fixture units, GPM flow rates for specialized equipment, natural gas load by both BTU/HR and CFH, CFH for air, vacuum & specialized gasses.
- Calculated data as required to select each piece of plumbing equipment including:
  - Hot water heater demand and storage capacity.
  - Sump pump sizing data.
  - Lift station sizing data.
  - Circulation pump sizing data.
- Pipe sizing data including: combined flow for each pipe run, developed length for each pipe run, selected pipe size based on stated criteria (sizing chart, pressure drop, fluid velocity, etc. This data may be either in tabulated form or rough pipe schematic.
- Rainwater leader calculations

Plumbing Working Drawings
General: Plumbing Working Drawings shall be sufficiently complete to assure high quality, fully functional plumbing systems, no contractual ambiguity, and also shall serve as a long term record of the building's plumbing systems. The following guidelines apply to all Plumbing

POTHOLING BEFORE?

- No work shall be called out in a manner which is not contractually enforceable. Plumbing quality level shall be fully identified to assure fair competitive bidding as well as a quality installation.
- Design / build format for the plumbing section of the work as is sometimes used in residential plumbing construction shall not be used on Cal Poly projects. An exception to this guideline shall be the case of the entire building project being design / build format when approved by the University's Representative.
- All points of connection to existing piping systems shall be identified.
- All piping shall be sized.
- The routing of all piping shall be indicated. The indicated routing shall be verified to be feasible within the furring spaces indicated in the architectural drawings.
- All equipment and fixtures shall be located. Maintenance access space as recommended by the manufacturer shall be indicated on the drawings.

California Polytechnic State University
Campus Facility Maintenance and Renovation Standards

- All piping penetrations of the building shell shall be indicated.
- Concrete core drills of existing structures shall be identified.

Provide a Plumbing Legend covering all symbols and abbreviations used in the plumbing drawings in order to have a fully enforceable contract.

Provide a Plumbing Schedule covering factory assembled plumbing equipment and fixtures. The intent should be the easy determination of the plumbing equipment included in the project to encourage competition among comparable product suppliers. Note that when a product manufacturer's name and model are called out on the schedule to serve as the basis of design, the specifications must be also be coordinated so that the same manufacturer is the first specified. See Section 01630; Product Options and Substitutions of Cal Poly's standard Division 1 for further information.

Provide Plumbing Floor Plans to scale for each level where plumbing services are to be provided. Provide enlarged partial plans for congested areas such as public rest rooms where the normal scale plans will not adequately depict the work.

Sections to scale should be provided when the vertical arrangement of equipment and piping relative to other building components is critical for proper function or adequate maintenance clearance, as well as to assure the feasibility of pipe routing through tight spaces.

Plumbing Riser Diagrams: Plumbing riser diagrams shall be provided for multi-story buildings whenever the relative configuration of piping components cannot be depicted in the plan view alone without ambiguity. As a minimum, riser diagrams shall be provided in the Plumbing Working Drawings for the following systems: all drain waste & vent systems including lab waste, all hot water systems with circulation, all pure water systems, and any other system which includes forced circulation by pumping. ✓

Plumbing Details: Plumbing details shall be provided as necessary to contractually assure a given level of quality. As a minimum, details shall be provided to cover all pipe penetrations of roofs, equipment supports, piping supports on roofs, backflow preventers, equipment installations where the relative position of plumbing appurtenances such as valves, unions, thermometer, drains, etc. are important to function, ease of service, and future replacement, pipe crossings of building expansion joints, pressure reducing assemblies, backflow prevention assemblies, roof drainage assemblies, floor drain assemblies, all pipe penetrations of exterior walls above and below grade.

Piping Diagrams (Schematics) shall be provided whenever the relative arrangements of plumbing components and equipment cannot be clearly depicted in the plans. As a minimum, piping diagrams shall be provided for in the Plumbing Working Drawings for water heating systems with circulation, pure water systems, and lift stations.

22 11 00 - Facility Water Distribution
Cal Poly owns and maintains the underground water distribution system throughout the campus.

Provide a domestic water pressure reducing valve and meter at each new building.

- CAN MULTIPLE METERS BE @ ~~[illegible]~~ SINGLE BLDG.
- IS THERE A STD METER?

Provide a separate water meter and RP backflow preventer for irrigation at each new building.

Fire water should not be metered or reduced in pressure. Back flow prevention is not required for fire water.

CHECK VALVES ~~ARE~~ REQ'D AT RISER? NOT NECESSARILY DDC?

All back flow preventers will by tested by a certified tester prior to the system by put in service

DCDA assemblies are required by Cal Poly and the County of S.L.O.

For new buildings to be connected to the campus water system, the anticipated additional water demand should be identified early during preliminary planning. This water demand should be submitted to the Principal Engineer for CAL POLY Physical Planning and Construction.

IS THIS POSITION STILL IN EXISTANCE.

Improvements to the campus system may be required to accommodate the additional demand. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

Design Conditions:
Water System Pressure: The water pressure assumed for system pipe sizing and design shall be the lower of the following:

- 60 PSI
- The actual system pressure.

Since the water pressure in the campus water mains varies throughout the campus depending on elevation within four water pressure zones, the Plumbing Designer should request the water pressure from the University's Representative for each project. The University's Representative will provide the known pressure at a given point (usually a fire hydrant). The Plumbing Designer will be responsible for determining the pressure at the point of use and should take into account the change in elevation between the known reference point and the point of use.

22 11 16 - Domestic Water Piping
Acceptable Piping Materials

- Above Grade: Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder for joining 2" and smaller pipe. 15% silver brazing conforming to AWS classification BCuP-5 for joining 2-1/2" and larger piping.
- Below Grade: Type K hard drawn or soft copper tubing, with wrought copper sweat fittings, all connections brazed with 15% silver brazing conforming to AWS classification BCuP-5 , wrap pipe with 1 layer of 10 mil tape. Soft copper tubing with radius bends shall be used to minimize below grade fittings. (Note: Pipe layouts which place water piping under the building slab should be avoided whenever other alternatives are feasible.)
- Below grade, beyond building footing, 3" and smaller piping, alternate material: Schedule 80 PVC pipe with socket fittings and solvent cement joints.
- Below grade, beyond building footing, 4" and larger: Cement-mortar lined Class 150 ductile iron pipe with bell and spigot joints and cast iron fittings. Specify restraining method: thrust blocks or restrained joints. Restrained joints should be called out for pipe configurations where thrust blocking will be difficult.
- Below grade, beyond building footing, 4" and larger, alternate material: Class 200 AWWA C-900 PVC with bell and spigot joints and cast iron fittings. Specify restraining method: thrust blocks or restrained joints. Restrained joints should be called out for pipe configurations where thrust blocking will be difficult.

IS EITHER OF THESE PREFERRED?

Water Pressure Reducing Valves
Provide a pressure reducing valve (PRV) which will limit the supply pressure to 80 PSI at the domestic water entry point to each building. This PRV shall be provided on water services with pressure below 80 PSI as well as above in order to provide additional protection in the event of a loss of pressure control in the campus water mains.

VALVE STD? MODEL #?

For low system pressures where a PRV may cause an undesirable pressure loss an exception to this requirement may be granted upon obtaining approval from the University's Representative.
Provide compound PRV's where required to accommodate a large range of flows.
Fire water should not be reduced in pressure. Separate fire water from domestic water ahead of the PRV.

Valving:
Water piping systems shall be provided with manual isolation valves at the following locations:

- At branch connections to underground system mains.
- At all building points of entry.
- On both sides of piped in devices which may need to be removed for servicing including water meters, back flow preventers, pressure reducing valves, strainers, pumps, etc. The devices shall be removable without draining down the building system.
- Exception: On small branch lines (1 inch and less) where the quantity of building water is small, the valve on the downstream side of the device may be omitted.
- At each floor where branch piping connects to a riser.

Domestic / Industrial Hot Water
For water conservation, all systems shall be designed to provide near immediate hot water at fixtures by using either a hot water circulation pump or electric heat tape. An exception to this is small systems with a short distance (50 feet or less) between the water heater and the fixture. Verify with the University Representative whether heat tape or circulation will be used on a project by project basis.

For systems with a hot water circulation, verify that the pump is not over selected and that the pipe velocities due to circulation will be reasonably low. (Copper fitting failure due to high velocity scouring has been a problem).

• ANY MAKE, MODELS?
• HIGH EFFICIENCY?

Grunfos Alpha Series Pumps

Domestic hot water heaters shall be gas fired, to minimize campus electrical demand as well as energy use cost.

TANKLESS?

Exceptions:

- Domestic hot water in buildings served by the campus central heating water system should be provided from a storage tank equipped with a double wall heat exchanger. (An exception to this may be allowed in the case of buildings with very low anticipated domestic hot water usage).
- Lavatories with small usage which are remotely located from a central system may be provided with hot water from a small electric storage type water heater with a maximum electrical demand of 1.5 kW. This exception shall not apply to shower usage. Instantaneous electric water heaters are not permitted due to high electrical demand.

Provide separate heating sources for domestic hot water and space heating water except for buildings served by the campus central heating water system.

Water heaters shall be installed in a sheet metal drain pan (smitty pan) with drain outlet piped to a safe location on the building exterior or floor drain.

Exception; Water heaters installed on a concrete floor slab where leakage would not damage the floor or building. Provide a floor area drain in the vicinity of the water heater.

All hot water piping shall be insulated. ✓ PRODUCTS ALLOWED?

Water systems shall be designed to prevent water hammer. Provide properly sized shock arrestors adjacent to all quickly closing valves including toilet flush valves, washing machines, dish washers, and solenoid valves. Provide a ball valve below each shock arrestor to allow for removal without system drain down. Provide access doors for all shock arrestors in concealed locations. Shock arrestor locations and sizes should be positively identified within the contract documents.

Disinfection

~~All potable water systems shall be disinfected and analyzed for bacteriological content. (Note:~~ On Cal Poly projects, fire protection systems should also be disinfected, since they are installed without approved backflow protection)

No fire system shall be installed without back flow protection

Bacteriological analysis shall be completed by a third party laboratory approved by the Cal Poly Office of Environmental Health & Safety (EH&S). The laboratory shall be submitted via the University's Representative for approval by EH&S a minimum of 72-hours prior to conducting disinfection. Analysis results shall be submitted via the University Representative to EH&S to certify compliance with the specifications. Disinfection procedure shall be repeated should EH&S indicate that compliance with the applicable regulations has not been achieved. ✓

Plumbing shop representative will need to be given a copy of the results before system can be put in service

Provide an industrial water system for non-potable uses at all wet laboratories, faucets with serrated tip for hose connections, dark rooms, and all other locations using toxic or hazardous materials and requiring water service for uses other than drinking. Backflow protection shall be by a reduced pressure principal type back flow preventer.

22 13 00 - Facility Sanitary Sewerage
Cal Poly owns and maintains the sanitary sewer system throughout the campus.

For new buildings to be connected to the sanitary sewer system, the anticipated additional sewer load should be identified early during preliminary planning. This load should be submitted to the Principal Engineer for Cal Poly Physical Planning and Construction. Improvements to the campus system may be required to accommodate the additional load. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

Systems discharging to the campus sewer system shall be in compliance with the requirements of the City of San Luis Obispo Waste Water Treatment Plant. Mechanical systems discharging unusual wastes may require special provisions or may not be allowed. Triggers include; high or low pH, oil, grease, chemical contamination, biological contamination, and rain water to the sewer. Mechanical systems typically impacted include: commercial kitchen waste, elevator sump pump discharge, lab process waste, and water purification system waste. The Design Professional shall be responsible for confirming impacts to the project due to the City Waste Water Discharge Ordinance. Alternatives for compliance shall be discussed with the University's Representative. Negotiations involving specific project issues shall be channeled through the Cal Polly Office of Environmental Health & Safety (Director).

For wet laboratories, the method which will be used for dealing with contaminated liquid wastes should be determined early in the design process in consultation with the University's Representative and Cal Poly Environmental Health and Safety (EH&S) . Contaminated liquid waste from laboratories are not allowed to be poured directly into drainage systems discharging to the campus sewers. EH&S should be consulted for procedures for dealing with liquid contaminated wastes. Contaminated wastes are typically containerized and held in the laboratory for disposal by EH&S as hazardous waste. Lab waste piping systems are provided in wet laboratories only for disposal of non-hazardous liquids and as a precaution in case of an accidental spill.

Soil and Waste Piping within a building and outside within five feet (5') of the foundation, except where indicated otherwise, shall be No-Hub cast iron pipe and fittings, asphaltum coated, free from defects, and shall conform to the requirements of CISPI Standard 301 ASTM A-888 or ASTM A-74 and manufactured by AB & I, Charlotte or Tyler. Fittings shall be up with Stainless Steel, Heavy-Duty No-Hub Couplings and shall be in compliance with ASTM C-1540 and ASTM C-564 Standards, except all above ground vent piping joints may be made up with Standard-Duty No-Hub Coupling in compliance with CISPI –310, and ASTM C-564 Standards.

Add Bulcher Pipe systems for FOG waste

LOCATION / FREQUENCY / ETC CLEAN OUTS?

Complete soil/waste drainage piping will be provided to each domestic plumbing fixture and will discharge by gravity. Sanitary drainage ventilation piping will be provided to each domestic plumbing fixture or trap and will terminate at various locations on the roof.

Horizontal sanitary, waste and vent piping will be designed for a uniform grade of 2% (¼" per foot).

IF NOT FEASIBLE IS THERE ALTERNATIVES?

HVAC condensate drainage piping (CD) will be provided to each separate HVAC unit, such piping will drain to an indirect waste connection to the sanitary soil/waste system via either tailpiece connection at the nearest lavatory or sink or indirectly to a floor sink.

Floor drains will be provided in laundry rooms and all toilet rooms. Drains will not be located under the partitions and will be provided with clearance for service work.

Underground cast iron pipe will be wrapped with 8 mil polypropylene pipe wrap to protect against low pH soil on campus.

All floor drains will be served by a trap primer. The trap primers will be flush-o-meter valve flushing tube or lavatory p-trap type. (No mechanical type trap primers will be provided).

Wall clean-outs at the end of runs for main lines ~~will~~ SHOULD BE provided, where possible.

Above rim height of highest fixture on that line on all floors

Shower drains will be fitted with removable hair screens within floor ~~cleanout~~ for ease of maintenance.

Drain

Sink drains will be cleanable without disassembly of the associated trap.

With cleanout above rim height of sink on all floors

Routing drainage piping overhead of electronic equipment, telecomm equipment and electrical equipment will be avoided.

Cleanouts will be the same nominal size as the pipe they serve; where they occur in piping eight inches and larger, six inches size will be used. All cleanouts will be accessible.

Cleanouts will be provided on all floors at the following locations:

- Horizontal offsets.
- End of soil, waste, water or storm drains more than five feet in length.
- Maximum of 50 foot intervals of horizontal runs within the building.

GREASE

above rim height of all sinks and lavatories on all floors
above rim height of all urinals
above/up stream all double santees and combinations.

- Base of vertical sanitary stacks and storm drain leader lines.
- Each change of direction if the total aggregate change exceeds 90 degrees.
- Main sewer connection outside of building. A 2-way cleanout with dual access plugs (threaded bronze, thermoplastic, or PVC) will be provided at this location. Cleanout will be installed in round cast iron valve box marked "Sewer".

Acceptable Piping Materials
Above Grade Sanitary Waste and Vent Piping: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, neoprene coupler with stainless steel clamp and shield.

SHOULD THIS BE AVOIDED WHEREVER POSSIBLE?

Below Grade Sanitary Waste and Vent Piping: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, extra heavy neoprene coupler with stainless steel clamp and shield (Husky, Clamp All, or equal)

Above Grade Vent Piping (Alternate materials)
- Schedule 40 galvanized steel pipe with recessed screwed cast iron drainage fittings.
- Type DWV copper with DWV solder fittings, 50/50 equivalent lead free solder.

Above / Below Grade Waste and Vent Piping (Alternate material for student housing buildings up to 3 stories only) Schedule 40 ~~ABS~~ DWV pipe and DWV fittings with solvent cement joints.

PVC

Waste vents shall be located a minimum distance of 30 feet horizontally from HVAC air intakes and occupied roof areas. Additionally, waste vents shall be extended with a support system to a level seven feet above the roof when the roof is occupied (such as in the case of roof mounted greenhouses). Waste vents shall not be located in confined roof wells where HVAC air intakes are located.

Provide accessible cleanouts at all levels of the building the same as required by code for first floor and lower levels.

Campus standard goes beyond code

Avoid sewage lift stations where possible. Gravity flow of sewage is preferred and is usually achievable due to the significant elevation change of most building sites on the campus. If a lift station is required, minimize the required capacity by segregating out upper level waste which can be gravity flow. Lift stations should be duplex pump type, powered by emergency power, with alarm contacts provided as an input to the building management system. Lift stations which handle rest rooms sewage shall be grinder type.

Floor drains shall be equipped with trap primers. Trap primer shall be accessible and equipped with a shut off valve to allow for servicing without system drain down.

22 14 00 - Facility Storm Drainage
Acceptable Piping Materials
Above Grade: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, neoprene coupler with stainless steel clamp and shield.

Below Grade Sanitary Waste: Service weight, no hub, cast iron soil pipe with cast iron DWV fittings, extra heavy neoprene coupler with stainless steel clamp and shield (Husky, Clamp All, or equal).

NO HDPE OR CONC.?

Insulate building storm water piping inside buildings to prevent building damage due to condensation on the pipe exterior.

22 15 00 - General Service Compressed-Air Systems
Acceptable Piping Materials;
Laboratory & Instrument Air Applications: Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing pre-washed, degreased and capped at both ends by manufacture. Fittings pre-washed, degreased and wrapped by the manufacturer. All joints brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to be prevent oxidation.

Alternate material for Shop Air Applications Only: Schedule 40 galvanized steel pipe with treaded 150 pound galvanized malleable iron fittings.

Alternate material for Shop Air Applications Only: Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder.

Building (house) compressed air shall not be shared with the building pneumatic compressed air system for HVAC controls. Building (house) compressed air and HVAC pneumatic control air shall be separate stand-alone systems complete with their own dedicated compressors.

For laboratory and instrument air, provide oil filter with automatic drain, and refrigerated air drier. For shop air applications, confirm requirements with users.

Valving: Compressed air systems shall be provided with manual isolation valves at the following locations:

- At each floor where branch piping connects to a riser.
- At the compressor.
- At branch connections to mains.
- Upstream of any equipment connected to compressed air.

Shop air systems typically require quick couple disconnect fittings for connection of flexible hoses. Confirm requirements with users.

Equipment:
Large house systems: Liquid ring type. (Nash or equal). Provide water saving circulation option with cooling water heat exchanger.

Requirements:
Drain Lines: Daylight to a conspicuous location. Plumb to sewer.
Hammer Arrestors: Install in a maintenance-accessible location for.

22 30 00 - Plumbing Equipment
Domestic Water System
Cold water systems will be sized using CPC flush valve curves, and hot water systems using flush tank curves. Equipment branches and main pipe size will be based on flow requirements without diversity. All plumbing valves will be located behind a lockable access panels when they are in a concealed location.

Hose bibs will be provided on the roof of each building. The hose bibs will be spaced 50 ft apart, maximum.

Chrome plated stops with gasket seats will be provided for sinks, lavatories, and wash basins when exposed to public view. A hose bibb with vacuum breaker will be provided under the lavatories in each toilet room. Exposed branch water supply piping in toilet rooms and custodial rooms will be chromium plated.

Water hammer arrestors will be provided in the wall, as required, behind a lockable access panel. Trap primers from lavatory tailpieces or water closet flush-o-meter valves for floor drains will also be provided. Isolation valves and unions at equipment connections will be provided.

The water piping will be designed to provide a minimum residual pressure at the most remote water closet of at least 30 psig. Pressure regulators will be installed to comply with California Plumbing Code.

- The maximum water pipe velocity will be 5 ft/sec. for hot water and 8 ft./sec for cold water.
- The minimum supply pipe size will be ½" for one plumbing fixture with a maximum flow of 1.25 gpm.
- Fixture pipe sizes will be as follows: 1-1/2" for a flush valve water closet, ¾" for a shower, ½" for sink and ¾" for hose bibs.

Domestic water piping within the buildings will be "Viega" pro-press or equal copper tubing and shall conform to ASTM B 75 or ASTM B88, Fittings shall be copper in conformance with ASTM B16.18, ASTM B16.22 or ASTM B16.26. O-rings for copper press fittings shall be EPDM.22 32 00 Domestic Water Filtration Equipment

Building Pure Water Systems

Building (house) pure water systems shall be designed to guarantee a water quality level of at least Lab Grade 3 per National Committee for Clinical Laboratory Standards (NCCLS) at all times. Individual laboratories requiring a guarantee of purer water shall install additional purification equipment in the laboratory as part of the building equipment (reference; Letter of Understanding dated 3-19-90, J. West; Physical Plant to K. McCaffrey; Natural Sciences).

The makeup water shall include pre-treatment consisting of a prefilter, a water softener, an activated carbon absorption filter, and a five (5) micron filter followed by treatment which shall consist of a reverse osmosis (RO) unit, mixed bed deionizer exchange service tanks, an ultra-violet light, and a 0.2 micron filter. Make up water capacity shall be determined in conjunction with sizing of the system storage tank.

The entire piping system shall be continually circulated. Dead legs shall be limited to drops to individual lab outlets. (Confirm maximum dead leg distance for each application) The entire circulating volume shall be exposed to ultra-violet light on each circulation. Provide dual stainless steel circulation pumps to allow for continued circulation in the event of pump failure.

The system shall include a storage tank sized to allow make up to occur over a 12 hour period while accommodating maximum anticipated use with 20 % excess capacity. Determination of maximum system use shall include a diversity factors which shall be determined with the University's Representative for each system.

The system shall include a side stream polishing loop which will continually circulate a portion of the water circulating through the system back through the mixed bed deionizers prior to return to the storage tank.

Provide a back pressure valve on the return piping to the storage tank set to assure adequate pressure at the highest most laboratory faucet.

Provide pressure gauges on both sides of all filters and pumps.

Building (house) pure water systems shall be monitored by the Building Management System to provide alarms to the Physical Plant watch stander at the campus Central Heat Plant. Alarms shall be generated in the case of Water quality (conductivity) out of range, low tank level, and no flow.

Acceptable Piping Materials:
Socket fused polypropylene prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system.

Schedule 80 PVC with solvent weld fittings prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system.

SCHEDULE OF MATERIALS?

Stainless steel could be used some applications

Confirm piping material with the University's representative on a case by case basis.

Valves shall be diaphragm type.

Valving:
Pure water piping systems shall be provided with manual isolation valves at the following locations:
At each floor where branch piping connects to a riser.

On both sides of piping elements which may need to be removed for servicing including filters, pumps, ultraviolet lights, mixed bed deionizers, and RO Units.

At branch connections to mains.
Provide provisions for system drain down to a floor drain including drain down of storage tanks.

22 35 00 - Domestic Water Heat Exchangers
Tube Bundles: Standard: Cupronickel Alloy. (Note: Standard grade copper tube bundles corrode within 3-5 years.)

Summary: Related Sections: Section 23 05 53.13 – Identification for Plumbing and HVAC Equipment.

Equipment: Water Heaters: 10 Year Tank Warranty; State, A.O. Smith, or equal

22 40 00 - Plumbing Fixtures
Lavatory

- Student Housing: Heavy duty commercial grade single handle faucet; deck mounted with 4" centerset, chrome plated finish, vandal resistant handle, brass body, washerless design, rotating ball control mechanism with replaceable nonmetallic seats and stainless steel socket, vandal resistant 0.5 GPM flow restrictors spray outlet, left rod operated metal drain.
    - Preferred Manufacturer and Models: Zurn: http://www.zurn.com/
        - Lavatory – Wall Hung: ~~Zurn Z5361~~, 20" by 18", high back.

American Standard Lucerne Model Number(s): 0356.028.020

    - Lavatory – Countertop: ~~Zurn Z5120 – Series, 19~~" round, vitreous china, self-rimming front overflow design.

American Standard Rondalyn Countertop Sink Model Number(s): 0490.156.020

  - Faucets: Preferred Manufacturer and Model: Moen CA83005 Non-mixing Electronic Faucet (0.325 GPM Vandal Resistant Flow). http://www.moen.com/search?search_scope=0&search_terms=CA8305
- Public: Heavy duty commercial grade single handle faucet; deck mounted with 4" centerset, chrome plated finish, vandal resistant handle, brass body, washerless design, rotating ball control mechanism with replaceable nonmetallic seats and stainless steel socket, vandal resistant 0.5 GPM flow restrictor spray outlet, and grid drain.
  - Campus Lavatory Standard (as appropriate):
    - 0.375 gpm Tamper Resistant Laminar Flow Control for Existing Lavatory faucet: Neoperl 40.7059.733
    - Moen 8413 F03 Comm 4" cc .35 GPM Lav Faucet w/o
    - Moen 8551 Sensored Lab Fct
  - Campus Basin Standard: Lavatory Basin with 4" Centerset Holes – American Standard Lucerne 0355.012.020
- Kitchen, Student Housing: Heavy duty commercial grade, single handle faucet, deck mounted with 8" centers, 8" swing spout, chrome plated finish, vandal resistant handle, brass body, rotating stainless steel ball control mechanism with replaceable non-metallic seats and stainless steel socket, vandal resistant 2 GPM flow restrictor aerator.

MODEL?

Shower Valves

- Student Housing: Institutional grade, pressure balancing, adjustable safety limit temperature stop control.
- Public: Institutional grade, pressure balancing, adjustable safety limit temperature stop control.
- Campus Standard: 1.5 GPM Low-Flow Pressure Compensating Traditional Showerheads – Delta 52634-15-BG

MODEL?

Fixture Traps:
Chrome plated, 17 gauge cast bronze.

Faucets, Supplies, and Trim:
Mechanical Trap Primers:
Standard: Reduce or eliminate use.

Hose Bibbs:
3/4" hose connection, chrome plated, with vacuum breaker, loose key.

Water Meters:
Domestic Water: Nutating disk, magnetic drive, bronze case

Irrigation Water: Turbine type, magnetic drive, bronze case

Pressure Reducing Valve: Bronze with Teflon disk and diaphragm Valves, General Purpose Ball Valves, 2-1/2" and smaller. Provide threaded valve with downstream union.

MODEL?
ANY DESIRE FOR REMOTE REPORTING?

22 42 13 - Commercial Water Closets, Urinals, and Bidets
Toilets

- Student Apartments:
  - Toilet shall be 1.28 GPF vitreous china round front bowl, with direct flow priming jet and reverse trap. Hydraulic performance exceeding ANSI A112.19.2 requirements.
  - Seat: All plastic, closed front with cover, stainless steel hinge.
  - Campus Preference:
    - 1.28 HET Floor Mount Floor Discharge – American Standard 2234.001.020 1.1 - 1.6
    - 1.28 HET Floor Mount Floor Discharge (ADA) – American Standard 3461.001.020 1.1 - 1.6
- Public Restrooms (including dormitories):
  - Toilets shall be 1.28 GPF, wall hung with support carrier, vitreous china, elongated, flush valve type exceeding ANSI A112.19.2 requirements.
  - Campus Preference:
    - American Standard 2257.101.020 with Afwall 1.1 – 1.6 (1.28)
    - 1.28 GPF HET manual Toilet (No Piston Valve) – Zurn Z6000 AV Model
    - 1.28 GPF HET Sensor Toilet (No Piston) – ZER6000 CPM Model
  - Flush valve: 1.28 GPF
  - Flush Valves - Toilets (with infrared sensors): ZER6000AV-ONE-CPM
    - Zurn ~~ZTS6000EV~~ high efficiency flushometer valve
    - Campus no longer installs Flush Valves - Toilets without infrared sensors.
  - Seat: All plastic, open front elongated without cover; stainless steel hinge.
- Alternate toilet for single stall public rest rooms where light use is expected: Toilet shall be 1.28 GPF with hydro-pneumatic pressure assisted flush, vitreous china, floor mounted, elongated bowl, exceeding ANSI A112.19.2 requirements. Seat: All plastic, open front elongated without cover; stainless steel hinge.
- Toilet carrier manufactures Zurn, Josam, Mifab and J.R. Smith
- Not Recommended:
  - Toilet: Zurn Z5615 Series, Zurn Z5655 Series, and Zurn Z5665 Series 1.28 GPF Elongated vitreous china toilet. The toilet bowls are too shallow for proper water scouring when flushed.
  - Toilet Seat: Enlongated Open Front Seat – Bemis 1955SSCT

Urinals
Campus Standard, as appropriate for use/scope:

- Washbrook 0.125 GPF FloWise Washout Top Spud Urinal, or
- *Z5758* "The Retrofit Pint" 0.125 gpf Ultra Low Consumption Urinal System (21" – 24" Footprint)
- Zurn-ZER 6003-CPM Model Sensor Operated Battery Powered Flush Valve for ¾" Urinals ZER6003AV-CPM

Trap Adapters:
Standard: Use when waste line has a tubular trap. Purpose is to allow for easier maintenance.
Preferred Manufacturer and Model: Arrowhead Brass Products, #6001A, 1-1/2" M.I.P. x 1-1/2" Tube Size – Brass Nut, Brass Ferrule with stop. Male in Female UPC-1003.2.
http://www.arrowheadbrass.com/products/

Flushometers:
Flush Valves: Standard: Zurn

22 45 00 - Emergency Plumbing Fixtures
Locations where hazardous materials are used or stored shall be provided with emergency eye washes in compliance with California Code of Regulations, Title 8 (CalOSHA) Section 5162, ANSI Z358.1 - 1990 and applicable ADA requirements.

Emergency eye washes and showers shall be supplied from the potable water system. Industrial water shall not be used to supply these fixtures.

Emergency eye wash and shower control valves shall remain open once activated until intentionally shut off.

Emergency eye washes and showers shall be located a distance no greater than 100 feet from the hazard and shall require no more than 10 seconds for an injured person to reach.

Floor drains shall ~~not~~ be provided at emergency showers in order to contain hazardous materials. The adjacent floor and walls should be designed to provide containment and also to be resistant to water damage. The plumbing design professional should coordinate this issue with the project Executive Architect.

22 47 00 - Drinking Fountains and Water Coolers
Water-Station Water Coolers:

- Hydration Station: Elkay Model LZWSSMSurface Mount bottle filling station.Sanitary, touchless activation with auto 20-second shut-off
- WaterSentry Plus 3000-gallon capacity filtration system, certified to NSF/ANSI 42 & 53 (Lead, Class 1 Partiulate, Chlorine, Taste 7 Odor)
- Integrated silver ion anti-microbial protection
- Fill rate: 1.5 gpm; 1.1 gpm when connected to a remote chiller
- Laminar flow to reduce splash
- Real drain system
- Visual user interface display includes:
- Green Ticker counts bottles saved from waste
- LED visual filter monitor indicates when to replace filter
- Optional remote chiller can be installed within 15 feet of unit to deliver chilled drinking water.

These are the two models that are currently being installed on campus With the Cal Poly X being the preferred unit of the campus community Elkay LZSTL8WSSP as the standard. Our retrofits end with an "X" for the CP model.

22 62 00 - Vacuum Systems for Laboratory and Healthcare Facilities
Acceptable Piping Materials
Type L hard drawn copper tubing with wrought copper sweat fittings, 95/5 tin antimony solder for joining 2" and smaller pipe. 15% silver brazing conforming to AWS classification BCuP-5 for joining 2-1/2" and larger piping.

Likely constituents of vacuum exhaust shall be determined and documented with the users to assist determination of vacuum exhaust location and degree of hazard due to discharged gasses.

Vacuum exhaust shall either be piped into the fume exhaust system or directly piped to a safe exterior location 7 feet minimum above the building roof. Vacuum exhaust discharge location should be carefully chosen so as not to create nuisance odors or hazards at operable windows, paths around the building, or other occupied exterior spaces. Vacuum exhausts which may create a nuisance or hazard should be piped to the same point as fume exhaust discharge for entrainment in the fume exhaust plume. House vacuum pump should be electrically interlocked with the fume exhaust fan in cases where the vacuum exhaust is directly connected to the fume exhaust system.

House vacuum pumps shall be protected by a drop out pot on the vacuum side of the unit.

Equipment:
Vacuum Pumps: Large house systems: Liquid ring type. (Nash or equal). Provide water saving circulation option with cooling water heat exchanger.

22 63 00 - Gas Systems for Laboratory and Healthcare Facilities
For Laboratory Buildings with a B Occupancy classification, verify that the quantity of stored gases classified as hazardous materials does not exceed the exempt quantities called out in the Building Code. H occupancy classification may otherwise be required.

Systems and storage for toxic gases shall be in compliance with the Uniform Fire Code including California Amendments in particular Section 8003, covering toxic compressed gases. Additionally, such systems shall be in compliance with the Toxic Gas Ordinance as written by the Western Fire Chiefs Association. Inform both the Executive Architect and University's Representative of requirements for gas storage cabinets, double containment piping, monitoring systems, and maximum quantity limitations in order to assure coordination among all disciplines.

Storage and piping concepts for laboratory gasses should be reviewed with Cal Poly Environmental Health and Safety via the University's Representative early in the design process. Issues include the possibility of asphyxiation (includes inert gasses) in confined spaces, as well as hazardous material concerns.

Laboratory gasses will not be used or stored in rooms having a HVAC system which returns air for recirculation.

Provide seismic restraint systems for all gas canisters and dewars.

Acceptable Piping Materials
Inert, Nontoxic Gases (nitrogen): Hard drawn ACR copper tubing with wrought copper sweat fittings. Tubing shall be pre-washed, degreased and capped at both ends by the manufacturer. Fittings shall be pre-washed, degreased and wrapped by the manufacturer. All joints shall be brazed with 15% silver brazing conforming to AWS classification BCuP-5. Brazing shall be done with nitrogen purge to prevent oxidation.

Other Gasses: Piping materials shall be selected on a project by project basis to meet specific program, compatibility, and regulatory requirements. Design for all toxic or hazardous gasses shall be reviewed and approved by EH&S.

Valving:
Laboratory gas systems shall be provided with manual isolation valves at the following locations:
At each floor where branch piping connects to a riser.
At the gas canister connections.
At branch connections to mains.

Upstream of any equipment connected to the laboratory gas system.

22 66 53 - Laboratory Chemical-Waste and Vent Piping
All wet laboratories, all fume hoods with cup sinks, and all other locations using chemicals which could damage standard DWV piping in the event of an accidental spill shall be provided with a lab waste and vent system. The lab waste and vent system shall be made of chemically resistant piping materials. The lab waste and vent system shall merge with the building sewer at a location outside of the building. Prior to combining the two systems, provide a lab waste sampling station (typically a manhole with a sump for sample collection).

Acceptable Piping Materials
Above and below grade Lab Waste and Vent Piping; (Preferred where permitted by code and allowed by Fire Marshal); Polypropylene acid resistant DWV piping and fittings conforming to ASTM D635 flammability requirements. Fusion welded fittings throughout except for trap connections, which shall be mechanical joint. (GSR Fuseal, Proline, or equal).

Above and below grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant coated cast iron pipe with matching mechanical joint DWV fittings. (Duriron or equal).
Above grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant glass acid piping with glass DWV mechanical joint fittings. (Pyrex , Kymax, or equal).

Charlotte Pipe STM F 2618: Standard Specification for Poly (Vinyl Chloride) (CPVC) Pipe and Fittings for Chemical Waste Drainage.
Scope: This specification covers the performance requirements of CPVC pipe, fittings and solvent cements used in chemical waste drainage systems.
F 493: Specification for solvent cements for Chlorinated Poly (Vinyl Chloride) (CPVC) plastic pipe and fittings.
Scope: This specification provides requirements for CPVC solvent cement to be used in joining CPVC pipe and socket fittings

Lab waste vents shall be located a minimum distance of 50 feet horizontally from HVAC air intakes and occupied roof areas. Additionally, lab waste vents shall be extended with a support system to a level seven feet above the roof when the roof is occupied (such as in the case of roof mounted greenhouses). Lab waste vents shall not be located in confined roof wells where HVAC air intakes are located. Provide accessible cleanouts at all levels of the building the same as required by code for the first floor and lower levels.

Floor drains shall not be provided in laboratories using hazardous materials as precaution against such materials being discharged to the sewer system. The floor system should be designed to provide containment of such materials in the event of an accidental spill.

Division 23 - Heating, Ventilating, and Air Conditioning (HVAC)
23 00 00 - Heating, Ventilating, and Air Conditioning (HVAC)
Code Rules and Safety Orders:
Construction as called out on working drawings shall be in full accordance with the rules and regulations of all ten parts of the California Code of Regulations, Title 24 (The California Building Standards Code) per University administrative policy. Title 24 includes amended versions of the Uniform Building Code, Uniform Mechanical Code, Uniform Plumbing Code, National Electric Code and the Uniform Fire Code. Also included is the California Energy Code (Title 24, Part 6).

Additional government codes and regulations shall be complied with as applicable to the project.

The regional air quality management district with jurisdiction over the Cal Poly Campus is the San Luis Obispo County Air Pollution Control District ((805) 781-5912) with headquarters in San Luis Obispo, California. Mechanical systems regulated by the district include: fume exhaust systems, cooling towers, fuel burning equipment (depending on size & fuel type), paint spray booths, incinerators, and dust handling equipment. A "Permit to Construct" must be obtained from the District prior to beginning construction or modification on any system regulated by the District. The Design Professional shall be the building). See applicable Sections for metering specifics. A complex of residential buildings may be master metered with approval of the University's Representative.

Design Documentation:
Design documentation shall be provided for all mechanical systems to assure efficiency in the mechanical design process and to serve as a reference after construction.

Conceptual Design Report / Mechanical Systems: At the Schematic Design phase, submit a "Conceptual Design Report" for the project's mechanical systems, covering each anticipated mechanical system in the project. A key element of the report shall be a discussion of the alternative systems available to meet the identified needs. Design Development work for the mechanical systems shall not begin until the University's Representative has approved a conceptual alternative for each mechanical system. The

Conceptual Design Report shall include:
- Identification of the mechanical services required in each type of space included in the project.
- The base criteria to be used for capacity sizing each mechanical service to each space type within the project. Identify the source of the criteria (e.g., consultant recommendation, program requirement, building user interview, code requirement, campus standard, equipment manufacturer). State extent to which services will be oversized to accommodate future growth.
- A discussion of the pro and cons of the design alternatives available for each mechanical system.
- A recommended alternative for each mechanical system.
- Space allocation required for each recommended alternative.
- A description of access to be provided to each major piece of mechanical equipment.

Design Development Report / Mechanical Systems:
The Design Development submittal shall include a "Design Development Report" covering each mechanical system in the project. This report shall include:
- Room by room documentation of required mechanical services.
- A written description of each mechanical system covering system purpose, system type, utility inputs, and locations of all major equipment.
- The criteria to be used in sizing and laying out each mechanical system and the source of the criteria (similar to Conceptual Design Report).
- Rough schematics of the proposed systems indicating major equipment, points of connection to existing utilities, and the rooms served by each mechanical utility.
- A description of the required space to accommodate each major element of the proposed mechanical systems, including mechanical rooms, exterior mechanical yards, roof mounted mechanical equipment, mechanical shafts, and ceiling space provisions.
- A complete description of access for maintenance and replacement to each piece of mechanical equipment.
- Rough equipment sizing calculations.
- Catalog cut sheets on all major mechanical equipment.

Title 24 Building Envelope Plan Check Compliance Documents: The Design Professional shall submit Title 24, Part 6; Building Envelope Plan Check Compliance Documentation no later than the 50% Working Drawing Submittal. This documentation shall include applicable portions of forms; ENV-1, ENV-2, & ENV-3.

- To support the stability of larger palms, consider backfilling with washed builder's sand to pack more easily and uniformly.
- There is no benefit to amending the backfill with organic matter. Use organic matter as mulch several inches deep and several feet out from the palm's base. Tamp out air pockets.
- If stabilization is required for large palms:
  - Use 2 x 4 or 4 x 4 wooden bracing attached against one-foot lengths of 2 x 4 vertically strapped or banded around the trunk.
  - Protect the trunk with nylon, burlap, or other suitable material where the one foot lengths of 2 x 4 are secured.
  - Do not nail into the trunk; nailing will cause permanent wounds, and disease and insect entry sites.
  - Palms may also be secured with guy wires or cable instead of wooden bracing.
  - Do not stabilize palms by planting them deeper in the hole; although some palms survive deep planting, most do not.
- Construct an irrigation berm four to six inches around the root ball and hole.
- Provide a two- to three- inch layer of mulch around the base of the palm to encourage new root growth, conserve moisture, and suppress weeds.
- Irrigate deeply and thoroughly.

Post-planting care

- Schedule irrigations based on need, not a clock or calendar.
- Irrigate sensibly, keeping the root ball and backfill evenly moist but not saturated.
- Keep turf grass and weeds away from the trunk base.
- Maintain a regular and complete fertilizer program once the palm is fully established.

<u>Division 33 - Utilities</u>

33 10 00 - Water Utilities

All underground plumbing utility piping shall have a minimum of 12" sand encasement.

All underground plumbing utility piping shall have a minimum depth of 24" unless otherwise approved by the administrative authority after all existing utilities have been potholed by the contractor.

DETAILS?

33 11 13 - Public Water Utility Distribution Piping

All underground domestic water piping 3" or less shall be schedule 80 PVC or HDPE SDR 11.

All underground domestic water piping 4" or larger shall be PVC C900.
All valves for underground domestic water 2" or larger shall be epoxy coated resilient wedge gate valves with 2" operating head.

THERE WAS A CAST IRON STD ALSO IN PREV. SECTION?

Preferred Manufactures: Mueller Co. or Clow Valve.

33 12 13.13 - Water Supply Backflow Preventer Assemblies

Preferred manufacturers are Wilkins and ~~Febco~~

I THINK IT IS NOW ZURN.

All backflow preventers shall meet AWWA standards.

STERILIZATION?

All backflow preventers installed by the contractor will be rested by a certified tester and documentation will be turned over to a campus representative before lines are put into service.

33 12 19 - Water Utility Distribution Fire Hydrants

Fire hydrant spacing shall comply with CFC and City of San Luis Obispo Fire Marshall's requirements.

All site building will be fully protected by fire sprinklers so a 75% reduction in fire flow can be applied with the authorization of the City San Luis Obispo's Fire Marshall.

Reduced pressure principle back flow preventers (RP devices) are required at any connection that would potentially contaminate the water supply. The RP device shall be a double detector check valve assembly at the connections to the fire risers and stand pipes.

All water pipes shall be installed with a minimum of 36" of cover.

Water meters shall be installed at every water service connection per the Plumbing Narrative.

Valves shall be arranged so that the system can be operated in a manner where no more than 5 fire protection apparatuses (hydrants, risers, stand pies, etc.) will be removed from service if repairs to the system are necessary.

Vertical clearance of all water pipes shall comply with State Department of Health separation requirements and water shall always be above the sanitary sewer.

Horizontal clearance shall comply with State Department of Health separation requirements with a minimum 10' separation of sewer and water measured from outside of pipe.

Fire hydrants shall be placed at a maximum of 300' from all portions of the buildings per City of San Luis Obispo Fire Marshall's requirements.

Fire department connections shall be placed a maximum of 40' from a fire hydrant per City of San Luis Obispo Fire Marshall's requirements.

Provide a minimum residual pressure at the roof elevation of 20 psi at fire flow (1,500 gpm) + average day demand (ADD).

Pipes less than 4" diameter shall be schedule 80 PVC. Or SDR11 HDPE

Pipes from 4" to 12" in diameter shall be C900 DR18 PVC.

All fittings shall be ductile iron.

All gate valves shall non-rising stem with resilient wedge seats.

All fire hydrants shall comply with City of San Luis Obispo Public Works Engineering Standards. requirements and have solid bronze bodies, shall be wet barrel, with one 2-1/2", and two 4-1/2" connections.

This standard is subject to change the City of SLO should be consulted

RP devices for fire connection shall be double detector check valves, with lockable OS&Y stems, leak detection meters, and shall employ reduced pressure principle check valves.

33 31 00 - Sanitary Utility Sewerage Piping
All sanitary sewer lines 4" or larger shall be SDR-35 PVC, lines 3" and smaller shall be DWV PVC or cast iron.

All manholes shall meet City of San Luis Obispo Public Works Engineering Standards.
Transition from existing clay and cast iron pipe to PVC shall be made with Fernco Rubber coupling.

POSSIBLY ADD DETAIL -

Sanitary Sewer Design Criteria
All sanitary sewer laterals and mains shall be installed with a minimum of 48" of cover. WHAT IF THIS NOT POSSIBLE?

Curved sewer mains shall be installed with a minimum curvature in compliance with manufacturer's recommendations. Joint deflections are not permissible.

All sanitary sewer mains shall have a minimum 8" diameter.

Sewer mains shall be designed with a minimum 2 fps and a maximum 10 fps half-full flow velocity and the following minimum slopes:
- 8" 0.50%
- 10" 0.35%

Force mains shall be designed with a minimum velocity of 4 fps and a maximum velocity of 8 fps.

Sewer laterals shall be 4" minimum diameter with a minimum slope of 2%.

Sewer mains shall have manholes at 300' maximum spacing, at angle points, at the beginning or ending of curves, clean outs may be used at the beginning of runs less than 150' long, otherwise a manhole is required.

Service laterals shall connect to the main with a wye pipe fitting and a clean out at the building connections and angle points. Laterals longer than 100' long shall require an additional clean out at approximately mid-point or at 100' maximum spacing.

Sewer pipes shall be installed a minimum of 12" below water lines in compliance with State Department of Health separation requirements.

Horizontal clearance between sewer and potable water mains shall be a minimum of 10' from outside of pipe in compliance with State Department of Health separation requirements.

Lift Station Design Criteria
Pumps shall be an alternating duplex system with individual pumps sized for PDD (250 gpm); Total dynamic head = ±120'. A triplex pumping system may be installed at the Design Builder's discretion and with University approval.

Accessories shall include an alarm system tied into a central monitoring station and odor control.

A stand by generator shall be included in the project to provide emergency power for the lift station. The generator must be sized to power both pumps in a duplex system or at least two pumps in a triplex system.

DIESEL OR NAT. GAS?

Sanitary Sewer Materials
All gravity mains and laterals shall be SDR 35 PVC.

All force mains shall be Schedule 80 PVC or C900 DR 14 PVC.

All Manholes shall be pre-cast concrete with eccentric cones, and traffic rated covers and frames.

CITY OF SLO?

Lift station wet wells shall be precast or cast in place concrete lined with a minimum 125 mils of type B polyurethane coating.

Lift station pumps shall be submersible pumps with oil filled, explosion proof motors with moisture sensors.

Pumps shall be mounted to a factory provided rail system

33 40 00 - Storm Drainage Utilities
Storm water management systems must meet RWQCB post-construction requirements as listed in the Post-Construction Storm water management Requirements for Development Project in the Central Coast Region (Resolution No. R3-2013-0032).

Project storm drain systems shall be designed for the 25-year design storm with 1' of freeboard at inlets, and overland relief shall be provided for the 100-year design storm.

Pipes less than 12" in diameter shall be SDR 35 PVC at a minimum of 0.5% slope; all pipe 12" or greater shall be dual wall type S Corrugated Polyethylene Pipe (CPP) with water tight joints at a minimum of 0.5% slope. All pipes shall have a minimum cover of 24". All pipes within the street shall be a minimum of 18" diameter.

Manholes or catch basins are required at angle points, at beginning or ending of curves, or on runs longer than 300'.

Provide minimum cover on all pipes in accordance with manufacturer's recommendations and not less than 12".

Provide a minimum of 12" of clearance to all other underground utilities.

Provide a minimum of 12" of cover from bio-retention swale gravel beds to top of pipes.

All manholes, catch basins, and junction boxes shall be concrete with cast iron lids or grates. All grates and lids shall be ADA compliant, bicycle friendly, and shall be traffic rated where there is a potential for vehicular traffic, including on sidewalks. Galvanized grates and lids are not allowed to prevent heavy metal contamination of storm water.

The bio-retention swale overflow inlets and drain pipe shall be designed to have adequate capacity to convey the 100-year storm with a minimum of 12" of free board in the swale.

Roof drains down spouts shall not directly connect to the storm drain system and the runoff should be directed to vegetated areas where possible. The intent of the storm drain system is to allow for bio-filtration of the storm water runoff and infiltration. Drainage should be conveyed in surface swale or by sheet flow where possible and pipes should be avoided unless necessary.

THIS HAS ISSUES IF NOT HANDLED CAREFULLY. MAY CONSIDER MORE INFO/REQS & POSSIBLY DETAILS? HOW CURBS, DRY WELLS ETC.?

Storage can be provided in high porosity aggregate or perforated pipes or storm chambers surrounded by high porosity aggregate. Storage can also be provided in the aggregate under porous pavers. The system should be designed to infiltrate completely in 48 hours or less. Excess runoff to these systems shall be connected to the storm drain collection system.

Contractors are responsible for preparing a Storm Water Control Plan (SWCP) in accordance with Post-Construction Storm water management Requirements for Development Project in the Central Coast Region (Resolution No. R3-2013-0032), to be approved by the RWQCB.

33 41 00 - Storm Utility Drainage Piping
All storm sewer lines 4" or larger shall be SDR-35 PVC, lines 3" and smaller shall be DWV PVC or cast iron.

All manholes shall meet City of San Luis Obispo Public Works Engineering Standards.

Transition from existing clay and cast iron pipe to PVC shall be made with Fernco Rubber coupling.

33 51 00 - Natural-Gas Distribution
~~Cal Poly owns and maintains the underground natural gas distribution systems throughout the~~ campus. Gas distribution pressure is 10 PSI. The campus also has a 60 PSI distribution system.

Commented [MD

This would be The Southeren California Gas Companies

5PSI

Provide a gas ~~regulator and m~~eter at each new building.

For new buildings to be connected to the campus natural gas system, the anticipated additional gas demand should be identified early during preliminary planning. This anticipated gas demand should be submitted to the Principal Engineer for Cal Poly Physical Planning and Construction. Improvements to the campus system may be required to accommodate the additional demand. The Principal Engineer shall identify a suitable point of connection to the campus system and what system improvements may be necessary to accommodate the new building.

33 51 13 - Natural-Gas Piping
All underground gas piping shall be polyethylene SDR-11 yellow and installed by a contractor/employee with current certifications in the type of fusion they are using.

All underground isolation valves for gas lines shall be polyethylene with 2-ich operating nut.

- MINIMIZE A.G. PIPING?
- VENT REQ'S?

Acceptable Piping Materials
- Above Grade 2" and smaller: Schedule 40 black steel pipe with treaded 150 pound black malleable iron fittings. Paint all exterior and exposed piping to prevent rust.
- Above Grade 2-1/2" and larger: Schedule 40 black steel pipe with welded steel fittings. Paint all exterior and exposed piping to prevent rust.

5PSI

Building Gas Pressure:
Regulate gas before entering buildings from ~~10 PSI~~ to 7 inches water column. Size the building gas distribution system based on a pressure of 7 inches water column. For some special high gas use applications such as large boilers, a 2 PSI mechanical room gas pressure may be allowed when approved by the University Representative and permitted by code.

Regulators:
Gas Regulators should be specified to include an under pressure shut off feature to automatically shut off the gas in the event of a downstream pipe rupture.

Earthquake Valves:
Provide seismic gas shut off valves on gas services to all buildings which are intended to be used for student housing or as a residence. Seismic gas shut off valves are not required on non-residential buildings.

PROXIMITY TO ROADS / OTHER SOURCES OF VIBRATION / INADVERTANT SHUT OFF

Gas Meters
- Gas meters may be either diaphragm bellows or turbine type as appropriate for the application. Gas meters shall record in cubic feet.
- Except where covered under c., below, provide valving and connections to allow for the use of a temporary meter bypass (3/4 inch hose) when removing the meter for servicing. This bypass should be configured such that the meter may be removed while still supplying limited gas service to keep pilot lights lit.
- For large academic buildings (greater than 30,000 gross square feet), commercial kitchens, and gas service which supplies an emergency generator, provide valuing and permanent piping for a full sized bypass of the meter. The bypass shall be configured to allow for meter removal without disruption of gas service.

METER STDS? MAKE / MODELS?

Gas piping shall not be routed under building floor slabs.

Prior to installing new gas appliances in existing buildings, verify the gas pressure with the University Representative. In some cases, the gas pressure has been increased to medium pressure to compensate for under sized gas piping. Provide an additional gas regulator near the appliance in the event that the existing building gas pressure exceeds the listing on the regulator to be furnished with the gas appliance. (Note: Gas appliances are often furnished with regulators which are only listed for a maximum inlet gas pressures of either 9 or 14 inches water column.) Provide regulator with a flow limiting orifice or with vent piped to outdoors as required by code. Review regulator location and safety issues with the University's Representative on a case by case basis. As a general rule, medium pressure gas should be reduced to low pressure prior to piping through spaces which are normally occupied. Medium pressure gas shall not be permitted in occupied spaces used for student housing or a residence.

Valving:
Gas piping systems shall be provided with manual isolation valves at the following locations:
- At building points of entry.
- At each floor where branch piping connects to a riser.
- At each gas appliance.
- At branch connections to underground system mains.

Gas meters to be installed with all boilers, generators, engine, etc., that may need a permit to operate, and that use natural gas, to allow for gas reporting requirements.